

STAT 5010

Statistical Methods and Applications II

FORMAT	CATEGORY	CREDITS
On-Campus	Core	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – Statistics Pathway (on-campus)	Machine Learning Engineer (47.3 / 100)	36.5 / 100
SPECIALIZATION Applied Mathematics Pathway		

36	78	+57
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Expands upon statistical techniques introduced in STAT 4000 . Topics include modern regression analysis, analysis of variance (ANOVA), experimental design, nonparametric methods, and an introduction to Bayesian data analysis. Considerable emphasis on application in the R programming language.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	47.3	90.1	+42.8	HIGH
Applied Scientist	42.8	99.7	+56.9	HIGH
Data Scientist	41.2	74.7	+33.5	HIGH
ML Engineer	35.2	68.0	+32.8	HIGH
AI Engineer	28.9	69.4	+40.4	HIGH
MLOps Engineer	23.3	67.0	+43.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning

1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	47.3 → 90.1 (+42.8 pts)
predictive learning tools exploratory development science models model	
Applied Scientist	42.8 → 99.7 (+56.9 pts)
biology computational research reasoning problems models learning real	
Data Scientist	41.2 → 74.7 (+33.5 pts)
toolkit presentation social problems	
ML Engineer	35.2 → 68.0 (+32.8 pts)
learning machine models model problems digital tools	
AI Engineer	28.9 → 69.4 (+40.4 pts)
models learning software model problems machine tools	
MLOps Engineer	23.3 → 67.0 (+43.8 pts)
model learning machine training tools models development	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.